

AE4262 Assignment

Instructions:

1. *Prepare a report on the assignment.*
2. *The report should contain at least the following parts*
 - *Problem description*
 - *Assumptions*
 - *Procedure*
 - *Results & observations*
 - *Conclusions & outlook*
3. *The report should be handed in **before 10th June 23:59 CET***
4. *You can do this assignment in a group of three/four students.*
5. *Please submit your assignment on Brightspace. If in a group, you still have to submit individually and indicate clearly name and student number of all authors at the beginning of your report.*
6. *Grading: the score in brackets represents the percentage of the grade that is assigned to each section. An additional 10% is assigned to the quality of the report.*
7. *Each team member is expected to write the full report and be fully aware of the topics discussed in it. If the instructors suspect that you copied the report without taking active part in it, your report might be invalidated and students would be reported to the board of examiners.*
8. *Where not specified, points are equally-distributed among subsections. Also, 10% of the total grade of this assignment is awarded for clarity, conciseness and presentation quality.*

CEA instructions: After downloading and unpacking the software, double-click on CEA executable (cea-exec). A subfolder will be created. Copy and past the input files on Brightspace within this folder. Edit the file as required by your exercise and save. Then run FCEA2m. A prompt will open: insert the exact name of your input file and press Return. An output file and an excel file will be created, containing the results of the computation.

CANTERA instructions: CANTERA is available in several programming languages, though the Python interface is recommended. Please note that the latest version (3.2) is not available in MATLAB. To install Cantera, use either Conda or Pip:

- `pip install cantera`
- `conda create --name ct-env --channel conda-forge cantera ipython matplotlib jupyter` *#Creates a Conda environment with CANTERA*
`conda activate ct-env` *#Activates the environment*

ipython, matplotlib and jupyter are optional. Python 3.10 or greater is required.

For further information refer to software's instructions (links provided on Brightspace and the tutorial's slides).

Experimental part

Please refer to the document “Laboratory Exercise_2026” for the description of the setup

Part 1 [5% of the final mark]

Flow calculations

- Write down and balance the global chemical reactions for methane-air and hydrogen-air combustion at stoichiometric, lean, and rich conditions, using a generic equivalence ratio ϕ .
- For each methane-air flames and hydrogen-air flames, show how the required fuel and air flow rates, flow velocities, and Reynolds numbers were determined. You should clearly show how the quantities needed for the Reynolds number calculation were obtained. Clearly state all assumptions made. Present your test matrices in a table.

Part 2 [15% of the final mark]

Flame speed calculations

- Calculate the laminar flame speed of the H_2 /air and CH_4 /air mixtures using the Bunsen flame angle method applied to the OH-PLIF images. Plot the measured flame speeds as a function of equivalence ratio. In the same plot, compare the experimental values with unstretched laminar flame speeds obtained from literature and from one-dimensional freely propagating flame simulations using Cantera. Discuss why the maximum flame speed does not occur at $\phi = 1$.
- Comment on any deviations between the experimental and theoretical values. Discuss possible sources of uncertainty in the flame-speed measurements and suggest improvements to the experiment and methodology to reduce errors.
- Discuss why the flame shape deviates from a perfect cone near the burner rim and at the flame tip.
- Explain why the hydrogen Bunsen burner had a smaller diameter than the methane Bunsen burner.

Part 3 [15% of the final mark]

Blowoff and flashback

- Discuss the physical mechanisms leading to blow-off and flashback in the hydrogen-air and methane-air premixed Bunsen flames considered in this assignment. Your discussion should consider flame speed, bulk flow velocity, equivalence ratio, heat losses, and boundary-layer effects.
- For the experimental burner geometries and thermal powers used in the experiment, estimate the fuel and air flow rates at which flashback is expected to occur. Compare these predicted values with the experimentally observed onset of flashback, and discuss possible reasons for any differences.

Part 4 [10% of the final mark]

Diagnostics

- Explain the differences in the flame emission spectra measured for the three premixed methane-air flames at different equivalence ratios ($\phi=0.8, 1, 1.2$) and identify the main chemical species responsible for the observed spectral emissions.
- Sketch the flame emission spectra expected for premixed hydrogen-air flames and diffusion methane flames. Use a graph with wavelength (200–800 nm) on the x-axis and relative emission intensity on the y-axis. Identify the main spectral features and emitting species expected for each flame type.
- For hydrogen-air flames, apply an Abel inversion to the OH* chemiluminescence images and compare the reconstructed fields with the OH-PLIF images for $\phi=0.7, 1, 1.3$. Discuss the differences between the OH* chemiluminescence and OH-PLIF measurements.

[Hint: you may use a one-dimensional flame simulation to compare the spatial profiles of OH concentration and heat release rate. Based on these profiles, where do you expect the OH* chemiluminescence signal to peak?]

Numerical part

Please refer to the tutorial slides on Brightspace for information on thermodynamics databases and use of the software. Sample exercises are also available on Brightspace.

Part 1 [15% of the final mark]

- Calculate and plot the variation of c_p for H_2 , O_2 , N_2 , CO_2 , H_2O and CH_4 from 300 to 2400K.
- The C_p for the gas mixture can then be calculated by using the relation

$$c_{p,mix} = \sum_{k=1}^N Y_k c_{p,k}$$

Calculate and plot (in the same figure) the variation of $c_{p,mix}$ for reactants and combustion products from 300 to 2400K, assuming the reactants to be air and fuel at equivalence ratios 0.5, 0.8, 1.0, 1.2, 1.4. Repeat the computations using hydrogen and methane as fuel. Assume complete combustion. What are your observations in terms of C_p variation (max 200 words)?

- Compute and plot the flame adiabatic temperature for complete combustion of H_2 /air and reactant temperature of 1100K for the same equivalence ratios. Compare this with the adiabatic flame temperature in the case of methane/air combustion.

What are your key observations and conclusions? Where do differences come from? (max 200 words)

Part 2 [20% of the final mark]

Chemical equilibrium (Using CANTERA or CEA)

Note: Use always baseline conditions unless indicated otherwise. Baseline conditions are indicated in **bold**

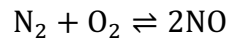
Note: Calculations have to be done at constant pressure and enthalpy (sample input files for NASA CEA are available on Brightspace)

Compute the following using hydrogen as fuel and then repeat the calculations using methane as fuel. Report the results for methane and hydrogen for each point below in the same graph.

- Evaluate the variation in adiabatic flame temperature with equivalence ratio for $\phi = 0.5, 0.6, 0.7, \mathbf{0.8}, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4$.
- Evaluate the variation in adiabatic flame temperature with Pressure, for **1**, 10, and 40 bar.
- Evaluate the variation in adiabatic flame temperature with air inlet temperature at **300K**, 600K, 1100K.
- Evaluate the NO emissions (in ppm) at 300K and 1bar for the range of ϕ in point a)

Part 3 [10% of the final mark]
Estimating NO_x

- a) [4 pts] Consider a two-steps reaction at 1 bar, where the first step is the global reaction of fuel/air combustion for $T_{\text{reactants}} = 300\text{K}$, and the second is the following equilibrium reaction:



Repeat the computation for both methane and hydrogen as fuel and for three equivalence ratios: $\phi = 0.5, 0.8$ and 1.0 . Plot the NO mass fraction as function of equivalence ratio.

Note: Only compute the composition in the first step. Use the equilibrium temperature from Part 2 for the second step.

- b) [6 pts] Now compare the results with those you computed numerically for both 1D flames and equilibrium, and comment on the differences. In particular explain in three separate bullet points (of max 50 words each)
- i) the difference between your NO computation and the one from equilibrium with CANTERA or CEA at lean conditions;
 - ii) the same at the stoichiometric condition; and
 - iii) the difference in NO obtained using chemical equilibrium (from CANTERA or CEA) and 1D flames.

AE4262 Assignment 2026 -- Annotated Technical Guide

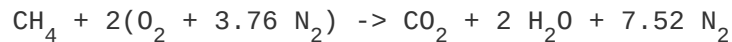
This document accompanies the original assignment PDF. Each section provides the technical methodology, governing equations, and insights extracted from lecture transcripts (20 May and 21 May 2026) and cross-referenced against the Warnatz combustion textbook (4th ed.), the Ulrich Maas lecture notes, and the Lecture Notes L1-L2 document. All formatting complies with the stated restraints.

EXPERIMENTAL PART

Part 1 -- Flow Calculations (5%)

1.1 Global Chemical Reactions

The balanced global reaction for methane-air combustion at a generic equivalence ratio ϕ is written by normalizing to one mole of fuel. For stoichiometric conditions ($\phi = 1$) the reaction is



[Lecture Notes L1-L2, Eq. 3.17-3.18] [Maas notes, Section 1, p. 5]

For a generic ϕ , the lean ($\phi \leq 1$) form is



[Lecture Notes L1-L2, Eq. 3.17]

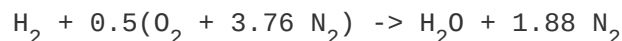
For rich conditions ($\phi > 1$) the limiting reactant becomes the oxidizer. Assuming complete combustion of the available oxygen



[Lecture Notes L1-L2, Eq. 3.18]

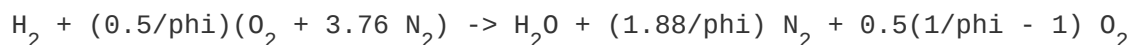
More precisely, for rich methane the products of complete combustion of the available oxygen would yield CO_2 and H_2O from the fraction $1/\phi$ of the fuel that reacts, with $(1 - 1/\phi)$ moles of fuel remaining. Note that in reality partial oxidation products such as CO and H_2 form under rich conditions, but the assignment specifies complete combustion for the flow calculation.

For hydrogen-air combustion at stoichiometry



[Maas notes, p. 5] [Lecture Notes L1-L2, Section 3]

At generic ϕ for lean hydrogen-air combustion



[Lecture Notes L1-L2, Eq. 3.17 applied to H_2]

1.2 Mass Flow Rate Derivation

All experiments operate at constant thermal power P_{th} . For methane, $P_{\text{th}} = 1.00 \text{ kW}$. For hydrogen, $P_{\text{th}} = 0.250 \text{ kW}$. The thermal power equals the mass flow rate of fuel that actually reacts multiplied by the lower heating value (LHV) of the fuel

$$P_{\text{th}} = \dot{m}_{\text{fuel,reacted}} * \text{LHV}$$

The LHV of methane is approximately 50.0 MJ/kg and of hydrogen is approximately 120.0 MJ/kg . The key distinction is which reactant limits the heat release.

For lean and stoichiometric conditions ($\phi \leq 1$), fuel is the limiting reactant and all fuel burns. Therefore $\dot{m}_{\text{fuel,reacted}} = \dot{m}_{\text{fuel}}$, giving

$$\dot{m}_{\text{fuel}} = P_{\text{th}} / \text{LHV} \quad (\phi \leq 1)$$

The air mass flow rate follows from the definition of equivalence ratio

$$\dot{m}_{\text{air}} = \dot{m}_{\text{fuel}} * s_{\text{air}} / \phi \quad (\phi \leq 1)$$

For rich conditions ($\phi > 1$), air (oxygen) is the limiting reactant. Only the fraction of fuel that the available oxygen can combust releases energy. To maintain the same P_{th} , the air flow must equal the stoichiometric air flow, because all available oxygen must react to release P_{th} of power. Thus $\dot{m}_{\text{air}}$ is held constant at its stoichiometric value

$$\dot{m}_{\text{air}} = P_{\text{th}} * s_{\text{air}} / \text{LHV} = \dot{m}_{\text{fuel,st}} * s_{\text{air}} \quad (\phi > 1, \text{ constant})$$

The fuel flow then increases linearly with ϕ

$$\dot{m}_{\text{fuel}} = \dot{m}_{\text{air}} * \phi / s_{\text{air}} = (P_{\text{th}} / \text{LHV}) * \phi \quad (\phi > 1)$$

For methane at 1 kW, $\dot{m}_{\text{fuel,st}} = 1000 / (50.0 \times 10^6) = 2.0 \times 10^{-5} \text{ kg/s}$. For hydrogen at 0.25 kW, $\dot{m}_{\text{fuel,st}} = 250 / (120.0 \times 10^6) = 2.083 \times 10^{-6} \text{ kg/s}$.

The stoichiometric mass-based air-to-fuel ratio s is defined as $s = (Y_{\text{Ox}}/Y_{\text{F}})_{\text{st}} = (\nu_{\text{Ox}} * W_{\text{Ox}}) / (\nu_{\text{F}} * W_{\text{F}})$. For methane, $s = (2 * 32) / (1 * 16) = 4.0$ on a pure-oxygen basis, and for combustion in air $s_{\text{air}} = 4.0 * (1 + 3.76 * 28/32) = 4.0 * 4.292 = 17.167$. For hydrogen, $s = (0.5 * 32) / (1 * 2) = 8.0$, giving $s_{\text{air}} = 8.0 * 4.292 = 34.333$.

[Lecture Notes L1-L2, Eq. 3.14-3.15] [Maas notes, p. 5]

1.3 Conversion to NLPM

The mass flow controllers report flow in normal litres per minute (NLPM). Normal conditions correspond to $T_n = 273.15 \text{ K}$ and $p_n = 1 \text{ atm}$ (1013.25 hPa). The density of air at normal conditions is $\rho_{n,\text{air}} = 1.293 \text{ kg/m}^3$. The conversion is

$$Q_n [\text{NLPM}] = (\dot{m}_{\text{dot}} [\text{kg/s}] / \rho_{n} [\text{kg/m}^3]) * 60000$$

[Laboratory Exercise 2026, Section 3.3] [Lecture transcript 20 May]

For the fuel MFCs, the normal density of the specific gas must be used. At normal conditions, $\rho_{n,\text{CH}_4} = p_{\text{W_CH}_4} / (R T_n) = (101325 * 0.016) / (8.314 * 273.15) = 0.7138 \text{ kg/m}^3$. Similarly $\rho_{n,\text{H}_2} = (101325 * 0.002) / (8.314 * 273.15) = 0.08921 \text{ kg/m}^3$.

The instructor emphasized the distinction between normal conditions (0 deg C) and standard conditions (20 deg C). Confusing them introduces approximately 7% error. The Bronkhorst MFCs used in this experiment operate on normal conditions.

[Laboratory Exercise 2026, Section 3.3] [Lecture transcript 20 May]

1.4 Reynolds Number Calculation

The Reynolds number of the pipe flow is $Re = \rho_{\text{mix}} * V_{\text{bulk}} * D / \mu_{\text{mix}}$. The mixture density at laboratory conditions ($T_{\text{lab}} = 300 \text{ K}$, $p = 1 \text{ atm}$) is computed using the ideal gas law

$$\rho_{\text{mix}} = p * W_{\text{mix}} / (R * T_{\text{lab}})$$

The mean molecular weight W_{mix} is determined from the mole fractions of fuel and air in the premixed stream. For a fuel with molar mass W_{F} entering at ϕ , the mole fraction of fuel is $X_{\text{F}} = 1 / (1 + \nu_{\text{O}_2} * 4.76 / \phi)$ and the remainder is air with $W_{\text{air}} = 28.97 \text{ g/mol}$.

[Maas notes, p. 5] [Lecture Notes L1-L2, Section 3]

The bulk velocity is $V_{\text{bulk}} = \dot{m}_{\text{dot_total}} / (\rho_{\text{mix}} * A)$, where $A = \pi * D^2 / 4$ is the burner cross-sectional area. For methane flames $D = 21 \text{ mm} = 0.021 \text{ m}$, giving $A = 3.464 \times 10^{-4} \text{ m}^2$. For hydrogen flames $D = 4 \text{ mm} = 0.004 \text{ m}$, giving $A = 1.257 \times 10^{-5} \text{ m}^2$.

[Laboratory Exercise 2026, Section 3.1]

The dynamic viscosity of the unburnt mixture can be approximated by the viscosity of air at 300 K, $\mu_{\text{air}}(300 \text{ K}) = 1.846 \times 10^{-5} \text{ Pa s}$. This is a reasonable approximation because air constitutes the overwhelming majority of the premixed stream by mass. The instructor confirmed in the lecture that specifying how each quantity in the Reynolds number was obtained is essential for full marks in this section.

[Lecture transcript 20 May, experimental part discussion]

The flow must remain laminar ($Re < 2300$ for pipe flow) to produce a stable conical Bunsen flame. If Re exceeds the critical value, the flow becomes turbulent and the classical cone-angle method for flame speed determination becomes invalid.

[Maas notes, Section 10.1, p. 44]

Part 2 -- Flame Speed Calculations (15%)

2.1 Cone Angle Method

The laminar flame speed S_L is determined from the Bunsen flame cone angle using the relation

$$S_L = V_u \sin(\alpha)$$

where V_u is the bulk velocity of the unburnt mixture at the burner exit and α is the half-angle of the flame cone. This relationship follows from the kinematic balance at the flame front where the component of the incoming flow velocity normal to the flame surface equals the laminar burning velocity.

[Laboratory Exercise 2026, Section 2.1] [Maas notes, p. 5] [Warnatz, Chapter on premixed flames] [Lecture 5 slides, p. 23-25]

The half-angle α is extracted from the OH-PLIF images. In these images the flame front is identified as the sharp transition between the region of negligible OH fluorescence (unburnt reactants) and the region of strong OH signal (post-flame zone). A straight line is fitted to the flame front on each side of the cone from the burner rim region (avoiding the tip and the base where deviations from conical geometry are largest), and the half-angle between this fitted line and the burner axis is α .

2.2 Why the Maximum Flame Speed Does Not Occur at $\phi = 1$

For both methane-air and hydrogen-air flames, the peak laminar burning velocity occurs at slightly rich equivalence ratios (approximately $\phi = 1.05$ for CH_4 and $\phi = 1.8$ for H_2) rather than at stoichiometry. Two effects contribute to this shift. First, the adiabatic flame temperature peaks near but slightly rich of stoichiometry because complete combustion releases the maximum heat when all oxidizer is consumed, but dissociation at high temperatures absorbs enthalpy. On the rich side, the excess fuel acts as a diluent with lower heat capacity than nitrogen, resulting in a higher equilibrium temperature compared to the same departure on the lean side. Second, chain-branching radical concentrations (H , O , OH) peak slightly rich of stoichiometry. The critical chain-branching reaction $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ has a high activation energy ($E_a = 70 \text{ kJ/mol}$ per the Maas notes, Section 6.1). Higher temperatures and optimal radical pools on the rich side enhance the overall reaction rate and thus the burning velocity.

[Maas notes, p. 29-31, Section 7.2 on flame velocities] [Lecture Notes L1-L2, p. 23-25 on adiabatic flame temperature] [Lecture 2 slides on T_{ad} peak shift] [Lecture transcript 20 May, instructor discussion on maximum flame speed]

2.3 Deviations from Perfect Cone Shape

Near the burner rim, the flame shape deviates from a perfect cone because of heat losses to the cold burner wall. The flame cannot propagate closer to the wall than the quenching distance δ_q , which for methane is approximately 2.5 mm and for hydrogen approximately 0.64 mm. Within this quenching distance, heat losses to the wall exceed the heat generated by chemical reactions, and the flame is locally extinguished. This creates a small gap between the base of the visible flame and the burner rim, and the flame front curves outward near the rim.

[Lecture 5 slides, p. 36-38, quenching distance table] [Maas notes, Section 9.2 on flame extinction at walls]

At the flame tip, the conical geometry produces strong positive curvature. The flame front is convex toward the unburnt gas, which enhances diffusive losses of heat and radicals from the reaction zone. Additionally, the two sides of the cone interact, and at the very tip the flame may open (for lean conditions) or close into a rounded cap depending on preferential diffusion effects governed by the Lewis number of the deficient reactant. For methane-air (Le approximately 1), the tip is approximately rounded. For hydrogen-air (Le approximately 0.3 for H_2), preferential diffusion of the highly mobile H_2 toward the curved tip locally enriches the mixture, raising the local burning velocity and causing the tip to sharpen or bulge.

[Lecture 8 slides, Lewis number table] [Maas notes, p. 29-31]

2.4 Why the Hydrogen Burner Has a Smaller Diameter

Hydrogen has a much higher laminar burning velocity than methane (approximately 200 cm/s at stoichiometry versus approximately 37 cm/s for methane). For a stable Bunsen flame, the bulk velocity of the unburnt mixture must exceed the laminar burning velocity, otherwise flashback occurs. The bulk velocity is $V = \dot{m} / (\rho \cdot A)$. At constant thermal power, the hydrogen fuel mass flow rate is much smaller (due to the high LHV of H_2) and consequently the total mixture mass flow is smaller. To achieve a sufficiently high bulk velocity to prevent flashback, the burner cross-sectional area must be reduced, hence the smaller diameter (4 mm for H_2 versus 21

mm for CH_4). Additionally, the quenching distance of hydrogen (0.64 mm) is much smaller than that of methane (2.5 mm), so a small-diameter tube can still sustain the flame without complete quenching.

[Lecture 5 slides, p. 33 and p. 38] [Laboratory Exercise 2026, Section 3.1] [Lecture transcript 20 May, instructor question on diameter difference]

2.5 Comparison with Literature and Cantera Simulations

The assignment requires plotting measured S_L from the cone-angle method alongside unstretched laminar flame speeds from the literature and from one-dimensional freely propagating flame simulations in Cantera. In Cantera, a FreeFlame object is created with the appropriate mechanism file (e.g., GRI-Mech 3.0 for methane, or a suitable H_2/O_2 mechanism such as the Li et al. or Burke et al. mechanisms). The freely propagating flame simulation solves the one-dimensional conservation equations for mass, species, and energy on an adaptive grid and returns the burning velocity as the inlet velocity at convergence.

Expected deviations between the Bunsen measurement and the 1D simulation include (a) flame stretch effects, since the Bunsen flame is not a flat unstretched flame and positive stretch at the tip or negative stretch near the base alters the local burning velocity, (b) heat losses to the burner rim, (c) non-uniformity of the velocity profile at the burner exit because the parabolic Poiseuille profile means V varies across the radius while the cone-angle method assumes uniform V_u , (d) three-dimensional effects not captured by the 2D cross-section, and (e) image resolution and noise in determining the cone angle.

[Lecture transcript 20 May, Part 2 discussion] [Lecture transcript 21 May, Cantera FreeFlame tutorial] [Maas notes, Section 7.2]

Part 3 -- Blowoff and Flashback (15%)

3.1 Physical Mechanisms

Flashback occurs when the flame propagates from the combustion zone back into the premixing tube. In the Bunsen burner configuration, the dominant flashback mechanism is boundary-layer flashback. Near the tube wall, the no-slip boundary condition forces the flow velocity to zero. At a distance δ_p from the wall (the penetration distance, which is slightly larger than the quenching distance δ_q), the flame can persist because the local burning velocity $S_f(\delta_p)$ exceeds the local flow velocity $u(\delta_p)$. The critical velocity gradient model of Lewis and von Elbe defines flashback onset as the condition $g_F < g_c$, where $g_F = (du/dy)|_{y=0} = \tau_w / \mu$ is the wall velocity gradient and $g_c = S_L / \delta_b$.

[Lecture 5 slides, p. 35-38, boundary layer flashback model] [Lewis and von Elbe, Combustion Flames and Explosions of Gases, Academic Press, 1987]

For fully developed laminar pipe flow, the wall shear stress is $\tau_w = 8 \mu V_{\text{bulk}} / D$, giving $g_F = 8 V_{\text{bulk}} / D$. Flashback occurs when $8 V_{\text{bulk}} / D < S_L / \delta_b$, that is when $V_{\text{bulk}} < S_L D / (8 \delta_b)$. Since hydrogen has S_L approximately 5 times that of methane and δ_q approximately 4 times smaller, the critical bulk velocity below which flashback occurs is much higher for hydrogen, making hydrogen flames far more susceptible to flashback.

Blow-off occurs when the bulk flow velocity greatly exceeds the laminar burning velocity. The flame lifts from the burner rim and is eventually extinguished because it can no longer anchor at the stabilization point. In the Bunsen configuration, blow-off is approached at very lean equivalence ratios (where S_L decreases) or at high flow rates. For hydrogen, blow-off is observed below $\phi = 0.5$, and for methane below $\phi = 0.8$, as indicated in the test matrices.

[Laboratory Exercise 2026, Table 1] [Lecture 5 slides, p. 33]

3.2 Estimating Flashback Flow Rates

To estimate the fuel and air flow rates at which flashback occurs, use the critical velocity gradient criterion. For a given equivalence ratio, compute S_L from the Cantera 1D flame simulation or from literature correlations. Approximate δ_b by the quenching distance (δ_b approximately equal to δ_q). Then the critical bulk velocity is $V_{\text{crit}} = S_L D / (8 \delta_b)$. The corresponding total mass flow rate is $\dot{m}_{\text{crit}} = \rho_{\text{mix}} * V_{\text{crit}} * A$. From $\dot{m}_{\text{crit}}$ and the known fuel-to-air mass ratio at ϕ , the individual fuel and air flow rates at the flashback limit follow. Compare these predictions to the experimentally observed flashback conditions reported in Table 1 of the Laboratory Exercise ($\phi > 1.55$ for H_2 and $\phi > 1.2$ for CH_4). Differences arise because the simple critical gradient model neglects (a) the temperature dependence of S_L near the wall, (b) heat losses to the wall that reduce the effective burning velocity, and (c) unsteady effects during the flashback transient.

[Lecture 5 slides, p. 37] [Lecture transcript 20 May, Part 3 discussion]

Part 4 -- Diagnostics (10%)

4.1 Emission Spectra of Premixed Methane-Air Flames

The spectrometer (Thorlabs CCS100, spectral range 350-700 nm) captures the visible emission from the flame. The key emitting species and their characteristic wavelengths are

Species	Emission Band (nm)	Transition
OH*	~306-310	$A^{2\Sigma} - X^{2\Pi}$ (UV, outside spectrometer range)
CH*	~431	$A^{2\Delta} - X^{2\Pi}$
C_2^*	~473, 516, 556	Swan bands $d^{3\Pi} - a^{3\Pi}$
CO_2^*	~350-600 (broadband)	Chemiluminescence continuum

[Lecture 6 slides, chemiluminescence section] [Lecture transcript 20 May, spectrometer demonstration]

At $\phi = 0.8$ (lean), the flame appears predominantly blue. The emission spectrum shows a strong CH* peak near 431 nm. C_2^* emission is weak or absent because C_2 radicals form primarily under fuel-rich conditions where incomplete combustion favors carbon-carbon bond recombination. The broadband CO_2^* continuum provides a low-level background.

At $\phi = 1.0$ (stoichiometric), the CH* peak remains prominent. A slight increase in C_2^* emission near 516 nm may become visible. The flame color remains blue.

At $\phi = 1.2$ (rich), the flame becomes greenish-blue due to the appearance of C_2^* Swan band emission at 473, 516, and 556 nm. The C_2 radical concentration increases because incomplete combustion under oxygen-deficient conditions allows acetylene (C_2H_2) and other C_2 species to accumulate. If the flame is sufficiently rich, soot particles form and emit broadband black-body radiation centered in the red/infrared region, giving the flame a yellow-orange luminosity. The instructor demonstrated this effect during the 20 May lecture, noting that the soot emission is continuous (not at discrete wavelengths) because it is thermal radiation from hot particles, fundamentally different from the chemiluminescence of gas-phase radicals.

[Lecture transcript 20 May, spectrometer demonstration and soot discussion] [Maas notes, Section 9.3 on PAH and soot formation]

4.2 Expected Spectra for Hydrogen and Diffusion Methane Flames

For premixed hydrogen-air flames, the emission spectrum in the visible range (350-700 nm) is expected to be nearly featureless. Hydrogen combustion produces no carbon-containing radicals (no CH*, no C_2^*) and no soot. The primary emitter is OH* near 306-310 nm, which falls outside the spectrometer range. Within the visible window, only weak water vapor (H_2O) rovibrational emission bands in the near-infrared tail may appear, plus a faint OH emission tail. The flame is essentially invisible to the human eye, which is a known safety hazard in hydrogen handling.

[Lecture transcript 20 May, hydrogen flame visibility discussion]

For a non-premixed (diffusion) methane flame, the emission spectrum features strong broadband soot luminosity (black-body continuum peaking around 600-700 nm) dominating over the discrete radical emission bands. The yellow-orange colour of a candle flame or a Bunsen burner with closed air ports is caused by this soot continuum. CH* and C_2^* peaks are still present but are superimposed on the much stronger broadband background.

[Lecture transcript 20 May, soot blackbody discussion] [Maas notes, Section 9.4 on soot formation]

4.3 Abel Inversion and OH-PLIF Comparison

OH* chemiluminescence images are line-of-sight integrated, meaning each pixel records the cumulative emission along the viewing direction through the entire flame volume. For an axisymmetric flame, the local (radial) emissivity distribution $\epsilon(r)$ can be recovered from the projected intensity $I(y)$ using the Abel inversion

$$\epsilon(r) = -(1/\pi) \int_r^\infty (dI/dy) / \sqrt{y^2 - r^2} dy$$

[Laboratory Exercise 2026, Section on Abel inversion]

After Abel inversion of the OH* chemiluminescence images, the reconstructed radial profile represents the local OH* emission rate, which is proportional to the local heat release rate. The OH-PLIF images, on the other hand,

provide a planar cross-section of the OH radical concentration distribution without line-of-sight integration (because the laser sheet selects a single plane).

Key differences between the Abel-inverted OH* and the OH-PLIF images arise because OH* chemiluminescence peaks at the reaction zone (where heat release is maximum and the formation rate of electronically excited OH is highest), while OH-LIF measures the ground-state OH concentration which extends well into the post-flame region (burned gas side). A 1D flame simulation comparing the spatial profiles of heat release rate and OH mole fraction confirms that heat release (and therefore OH* emission) peaks sharply at the flame front, while the OH concentration profile has a broader distribution extending downstream.

[Lecture transcript 20 May, Abel transform and PLIF vs chemiluminescence discussion] [Lecture 6 slides, chemiluminescence section] [Maas notes, p. 67, OH distribution in jet flame]

NUMERICAL PART

Part 1 -- Thermodynamic Properties (15%)

N1.1 Specific Heat Capacity Cp of Individual Species (Part a)

The specific heat at constant pressure for each species is computed using the NASA (JANNAF) polynomial fits. For each species, the molar Cp is

$$C_p/R = a_1 + a_2 \cdot T + a_3 \cdot T^2 + a_4 \cdot T^3 + a_5 \cdot T^4$$

where $R = 8.314 \text{ J/(mol K)}$ is the universal gas constant and a_1 through a_5 are polynomial coefficients tabulated for two temperature ranges (typically 200-1000 K and 1000-6000 K). These coefficients can be obtained from the NASA thermodynamic database (searchable online as 'NASA polynomials') or from the mechanism file used in Cantera (e.g., GRI-Mech 3.0 includes thermodynamic data for all species).

[Lecture Notes L1-L2, Section 3 on JANNAF polynomials] [Lecture transcript 20 May, numerical Part 1 discussion] [Lecture transcript 21 May, Cantera mechanism file tutorial]

The species to plot are H_2 , O_2 , N_2 , CO_2 , H_2O , and CH_4 from 300 K to 2400 K. Observations expected from this plot include that the Cp of diatomic molecules (H_2 , O_2 , N_2) increases modestly with temperature due to the activation of vibrational degrees of freedom. Triatomic and polyatomic molecules (H_2O , CO_2 , CH_4) have higher Cp values and stronger temperature dependence because they possess more vibrational modes.

[Lecture transcript 21 May, Cp discussion and temperature dependence]

N1.2 Mixture Cp (Part b)

The mixture specific heat on a mass basis is $C_{p,\text{mix}} = \sum_i (Y_i \cdot C_{p,i})$. The mass fractions Y_i depend on the equivalence ratio and whether the mixture consists of reactants or products. For reactants at a given ϕ , the mass fractions follow directly from the balanced global reaction (Section 1.1). For products, assuming complete combustion, the product composition is determined by the same balanced equations.

[Lecture Notes L1-L2, Eq. 3.9] [Lecture 2 slides, mixture Cp]

The instructor noted that the Cp of the product mixture is generally higher than that of the reactant mixture at the same temperature because the products contain triatomic species (CO_2 , H_2O) with more vibrational modes than the dominant reactant species (N_2 , O_2).

N1.3 Adiabatic Flame Temperature at $T_{\text{reac}} = 1100 \text{ K}$ (Part c)

For complete combustion, the adiabatic flame temperature is found by solving the enthalpy balance $h_{\text{prod}}(T_{\text{ad}}) = h_{\text{reac}}(T_{\text{reac}})$. Using the simplified form from the Lecture Notes L1-L2

$$T_{\text{ad}} = T_0 + (\Delta H_R \cdot Y_{F,r}) / (C_{p,\text{prod}} \cdot \nu_F \cdot W_F) \text{ for } \phi \leq 1$$

$$T_{\text{ad}} = T_0 + (\Delta H_R \cdot Y_{Ox,r}) / (C_{p,\text{prod}} \cdot \nu_{Ox} \cdot W_{Ox}) \text{ for } \phi > 1$$

[Lecture Notes L1-L2, p. 23-26] [Lecture 2 slides, adiabatic flame temperature derivation]

Key observations from comparing H_2/air and CH_4/air at $T_{\text{reac}} = 1100 \text{ K}$ include that (a) H_2 produces a higher adiabatic flame temperature than CH_4 at stoichiometry (approximately 2380 K for H_2/air versus 2222 K for CH_4/air at 298 K inlet, with an increase when $T_{\text{reac}} = 1100 \text{ K}$), (b) the peak shifts slightly rich of $\phi = 1$ due to dissociation effects (discussed in Section 2.2), and (c) at elevated reactant temperatures the gain in T_{ad} is less than the increase in T_{reac} because dissociation is enhanced at higher product temperatures (the Le Chatelier principle shifts the equilibrium toward the endothermic dissociation side).

[Maas notes, p. 18, adiabatic flame temperature examples] [Lecture transcript 21 May, discussion on T_{reac} effect and dissociation]

Part 2 -- Chemical Equilibrium with Cantera or CEA (20%)

N2.1 Workflow in Cantera

In Cantera (Python interface), the equilibrium calculation proceeds as follows. First, create a gas object from a mechanism file (e.g., GRI-Mech 3.0 for methane, or a hydrogen mechanism). Set the initial state by specifying temperature, pressure, and composition. Then call the `equilibrate('HP')` method to solve for chemical equilibrium at constant enthalpy and pressure, which yields the adiabatic flame temperature and the equilibrium species composition.

```
The essential Cantera calls are as follows. Create a Solution object from
'gri30.yaml'. Call set_equivalence_ratio with the desired phi, fuel string, and
oxidiser string (for air, specify O2 and N2 in the ratio 1 to 3.76). Assign the
tuple (T_reac, P) to gas.TP. Call gas.equilibrate('HP'). The adiabatic flame
temperature is then gas.T and the NO mole fraction is gas['NO'].X[0].
```

[Lecture transcript 21 May, Cantera equilibrium tutorial]

The instructor emphasized that when specifying composition, mole fractions must be used (not mass fractions) unless the 'basis' is explicitly changed. The default normalization ensures fractions sum to unity.

N2.2 Baseline Conditions and Parameterization

All calculations use baseline conditions unless stated otherwise. Baseline conditions are $T_{\text{reac}} = 300 \text{ K}$ and $p = 1 \text{ bar}$. The equivalence ratios for part (a) span $\phi = 0.5$ to 1.4 in steps of 0.1 . Part (b) varies pressure to $1, 10, 40 \text{ bar}$ at $\phi = 1.0$ only. Part (c) varies inlet temperature to $300, 600, 1100 \text{ K}$ at $\phi = 0.8$ and $p = 1 \text{ bar}$. Part (d) reports NO mole fraction in ppm ($X_{\text{NO}} \times 10^6$) at 300 K and 1 bar over the full range of ϕ .

[Assignment 2026, Numerical Part 2] [Lecture transcript 20 May, baseline conditions clarification]

For each sub-part, results for H_2 and CH_4 must appear on the same graph for direct comparison.

N2.3 Physical Interpretation of Pressure and Temperature Effects

Increasing pressure increases the adiabatic flame temperature because dissociation reactions generally increase the total number of moles (e.g., $\text{CO}_2 \rightarrow \text{CO} + 0.5 \text{ O}_2$ increases moles by 0.5). By Le Chatelier's principle, higher pressure shifts the equilibrium toward the side with fewer moles, suppressing dissociation and retaining more enthalpy in the products as sensible heat rather than chemical potential energy. At 40 bar the T_{ad} increase relative to 1 bar is on the order of $30\text{-}50 \text{ K}$.

[Maas notes, p. 18] [Lecture transcript 21 May, pressure effect on T_{ad}]

Increasing reactant temperature also raises T_{ad} but by less than the $\Delta(T_{\text{reac}})$ increment because (a) more dissociation at higher product temperatures absorbs enthalpy, and (b) the heat capacity of the products increases with temperature. The instructor stressed that this is a thermodynamic manifestation of Le Chatelier's principle applied to the overall exothermic combustion process.

[Lecture transcript 21 May, reactant temperature effect and Le Chatelier discussion]

Part 3 -- Estimating NO_x (10%)

N3.1 Two-Step Calculation Procedure (Part a)

The calculation proceeds in two sequential steps. Step 1 is the global combustion reaction for fuel/air at $T_{\text{reac}} = 300 \text{ K}$, assuming complete combustion. This yields the product composition (major species only) and the adiabatic flame temperature from Part 2. Step 2 treats the equilibrium reaction $\text{N}_2 + \text{O}_2 \rightleftharpoons 2 \text{NO}$ occurring in the post-combustion gas at the equilibrium temperature obtained in Part 2.

[Assignment 2026, Numerical Part 3a] [Lecture transcript 20 May, NO_x procedure]

The equilibrium constant K_p for $\text{N}_2 + \text{O}_2 \rightleftharpoons 2 \text{NO}$ is

$$K_p = \exp(-\Delta G_R / (R T))$$

where ΔG_R is the standard Gibbs free energy of reaction at temperature T . This can be computed from $K_p = (X_{\text{NO}}^2) / (X_{\text{N}_2} \cdot X_{\text{O}_2}) \cdot (p/p_0)^{(2-1-1)} = X_{\text{NO}}^2 / (X_{\text{N}_2} \cdot X_{\text{O}_2})$ since the sum of stoichiometric coefficients is zero ($\Delta \nu = 2 - 1 - 1 = 0$), and the pressure term vanishes.

[Lecture Notes L1-L2, Sample Exercise 1 on equilibrium constant] [Maas notes, p. 18, equilibrium constant definition]

The instructor clarified that the first step only gives the composition of the major species (CO_2 , H_2O , N_2 , and excess O_2 or fuel). These species form the background mixture in which the NO equilibrium is computed. It is not simply two-species equilibrium. The presence of water vapor and other species affects the mole fractions of N_2 and O_2 available for the reaction. The calculation must be repeated for $\phi = 0.5, 0.8$, and 1.0 for both CH_4 and H_2 (six total calculations).

[Lecture transcript 20 May, Part 3 clarification]

N3.2 Comparison and Discussion (Part b)

At lean conditions ($\phi = 0.5$ and 0.8), the two-step hand calculation will predict lower NO than the Cantera/CEA equilibrium because the hand calculation uses only the global reaction followed by a single equilibrium reaction, neglecting all minor species and their interactions. The equilibrium solver considers hundreds of reactions simultaneously, including the formation of O, H, OH radicals that affect the $\text{N}_2/\text{O}_2/\text{NO}$ balance. At lean conditions, excess O_2 favors NO formation, and the Cantera equilibrium produces more radicals that participate in additional NO pathways.

At stoichiometric conditions ($\phi = 1.0$), the difference between the two-step calculation and Cantera/CEA equilibrium is smaller because the product temperature is highest and the simple thermal NO model (which is temperature-dominated) captures the dominant physics. However, at stoichiometry there is essentially no excess O_2 from complete combustion, so the hand calculation may under-predict NO because it does not account for the O_2 produced by dissociation of CO_2 and H_2O at high temperatures.

The difference between Cantera/CEA equilibrium NO and the NO from a 1D flame simulation arises because NO formation is kinetically controlled, not equilibrium-controlled. The thermal NO mechanism (Zeldovich mechanism) has a very high activation energy ($E_a = 319 \text{ kJ/mol}$ for the rate-limiting step $\text{O} + \text{N}_2 \rightarrow \text{NO} + \text{N}$), making the formation rate extremely slow compared to the residence time in a flame. In a 1D freely propagating flame, the gas passes through the high-temperature zone in a fraction of a millisecond, which is insufficient for NO to reach equilibrium. Consequently, the 1D flame simulation predicts substantially less NO than the equilibrium calculation. The equilibrium solver overpredicts NO because it assumes infinite time for all reactions to reach their final state.

[Maas notes, Section 8.1 on thermal NO, p. 33-34] [Lecture transcript 20 May, NO_x comparison discussion] [Lecture transcript 21 May, kinetic vs equilibrium NO]

Each of the three bullet-point answers in Part 3b must be concise (maximum 50 words each) as specified in the assignment instructions.

Additional Guidance from Lecture Transcripts

Cantera 1D Flame Simulation Procedure

The 21 May tutorial provided the following workflow for the 1D freely propagating flame simulation. A gas object is created from the mechanism file. The initial composition is set at the desired ϕ . A FreeFlame object is constructed with an initial domain width (e.g., 0.04 m) and an initial grid. The flame is solved iteratively, starting with a coarse grid and loose tolerances, then refining. The instructor recommended using the auto refinement feature by calling `flame.solve(loglevel=1, auto=True)` which progressively tightens the grid and convergence criteria. The laminar burning velocity is obtained as `flame.velocity[0]` (the inlet velocity at convergence). Species profiles are accessed via `flame.X` (mole fractions) or `flame.Y` (mass fractions) as functions of the axial coordinate `flame.grid`.

[Lecture transcript 21 May, Cantera FreeFlame tutorial section]

Report Quality Requirements

The instructor emphasized that the report must be a concise engineering document. Answers should be direct and avoid circular or verbose discussions. The grading allocates 10% to clarity, conciseness, and presentation quality. Each team member must understand and be able to justify all parts of the report, regardless of internal task division. Use of AI tools is permitted for improving readability, but the technical content must reflect genuine understanding. The deadline is 10 June 2026, 23.59 CET. Submission is via Brightspace, individually.

[Lecture transcript 20 May, assignment instructions section]

Key Thermodynamic Relations

The absolute enthalpy of species i is $h_i = \Delta h_{f,i} + \int_{T_0}^T C_{p,i} dT$. For the adiabatic flame temperature calculation, the enthalpy balance is $\sum_i Y_{i,r} h_i(T_r) = \sum_i Y_{i,p} h_i(T_{ad})$. The JANNAF polynomial coefficients provide $C_p(T)/R$ as a 4th-order polynomial in T . Two sets of coefficients exist for each species, valid in the low-temperature range (typically 200-1000 K) and the high-temperature range (1000-6000 K). The coefficients are stored in Cantera mechanism files in YAML format or in the older Chemkin-II format.

[Lecture Notes L1-L2, p. 22-23] [Lecture transcript 21 May, NASA polynomial and mechanism file discussion]

Le Chatelier's Principle in Combustion Context

The instructor discussed Le Chatelier's principle extensively during the 21 May lecture. For the dissociation reaction $\text{CO}_2 \rightleftharpoons \text{CO} + 0.5 \text{O}_2$, increasing pressure shifts equilibrium toward CO_2 (fewer moles side), suppressing dissociation and increasing T_{ad} . Increasing temperature shifts equilibrium toward the endothermic direction (dissociation), which is why the gain in T_{ad} from higher T_{reac} is sublinear. For the $\text{N}_2 + \text{O}_2 \rightleftharpoons 2 \text{NO}$ reaction ($\Delta n = 0$), pressure has no effect on the equilibrium constant because the total moles do not change. However, the NO formation rate increases with pressure because the concentrations of N_2 and O_2 increase.

[Lecture transcript 21 May, Le Chatelier discussion] [Maas notes, p. 36 on N2O formation and pressure dependence]